# The Empirical Model in *photic*: Lyzenga's Log-Linear Fit, Diffuse Attenuation and Depth Errors

Technical summary

20 October 2026

**Abstract**

`MODEL lyzenga_log_linear` fits Lyzenga's log-linear depth model to soundings, such as ICESat-2 bathymetric photons. The model can be fitted to one image or to several scenes at once. This note sets out how *photic* makes three estimates.

1. The depth, and its $1\sigma$ prediction error (`ERROR`), from a robust weighted least-squares fit.
2. The diffuse attenuation coefficient $K$ of every scene and band, as a smooth field over the scene (`ATTENUATION`, stage 1). It is estimated at the soundings, using the physical model that Lyzenga's regression linearises.
3. A physical depth at every cell, from how the attenuation differs between bands and scenes, with its own error estimate (stage 2).

All of it is implemented in C, in `model/lyzenga.c`. The results quoted are from the Arabian Gulf validation site between Dubai and Abu Dhabi: six Landsat 8 scenes, against ICESat-2 ATL24 split by pass into calibration and test halves.

Lyzenga's fit across all scenes jointly scores 90.0% hits on the test half. Its error estimate holds 80% of the errors within $\pm 1\sigma$, but its tails are heavier than Gaussian. $K$ at 443–561 nm is plausible and spatially coherent. $K$ at 655 nm falls below pure water's absorption, so it is not physical. The depth from the attenuation is a diagnostic, not a product, and its error estimate is too small by a factor of about two to three.

## Contents

**1 Notation and inputs** **2**

**2 Lyzenga's regression** **2**
2.1 The model and its fit . . . 3
2.2 Rescaling against regression dilution (`POWER`) . . . 3
2.3 The depth error estimate (`ERROR`) . . . 3

**3 The attenuation behind the regression** **4**

**4 Stage 1: the attenuation coefficients at the soundings** **6**
4.1 A smooth field for each grid . . . 6
4.2 Weights from the noise . . . 6
4.3 Selecting soundings by the fitted signal . . . 7
4.4 Outputs . . . 7

**5 Stage 2: a depth from the attenuation** **7**
5.1 A straight line at every cell . . . . . 7
5.2 The depth error estimate . . . . . 8

**6 Results at the Gulf** **8**
6.1 The attenuation coefficients . . . . . 8
6.2 The depth from the attenuation . . . . . 9

**7 Usage** **9**

# 1 Notation and inputs

**Grids.** The command takes $n$ band grids $L_m$, $m = 1, \ldots, n$, all on one grid. They can be remote-sensing reflectance from ACOLITE, or radiance. With `ATTENUATION ... SCENES` $S$, the grids are scene-major: $S$ scenes of $n_b = n/S$ bands each, in the same band order. Grid $m$ belongs to scene $s(m) = \lfloor m/n_b \rfloor$.

**Window means.** Each band is averaged over a $(2r-1) \times (2r-1)$ window of valid cells (`RADIUS` $r$), giving $\bar{L}_m(\mathbf{x})$. A cell is modelled only where every band is valid. `LAND` and `SHALLOW` masks, if given, further restrict it.

**Deep-water signal and noise.** For each band:

- $L_{\infty,m}$ is either given (`LSM`) or the $p$-th percentile of the window means (`PERCENTILE`, default $p = 1$).
- $\sigma_m$ is the robust spread of the darkest cells: 1.4826 times the median absolute deviation of the window means at or below that percentile. It stands for the noise of a deep-water cell.

The signal above deep water is

$$v_m(\mathbf{x}) = \bar{L}_m(\mathbf{x}) - L_{\infty,m}. \tag{1.1}$$

A band sees the bottom while $v_m \geq k\,\sigma_m$ (`DEEP` $k$, default 2). A cell where any band fails this test is optically deep for the Lyzenga fit, and gets no depth.

**Soundings and tides.** Soundings $z$ are depths below the datum, positive down. Each sounding goes to its nearest cell; only soundings within `DEPTHS` $[z_{\min}, z_{\max}]$ are used. `TIDE` $h$ is the water level above the datum when the image was taken, so the light saw a water column $Z = z + h$. With several scenes, scene $s$ has its own level $h_s$ (`TIDES`), and $Z_s = z + h_s$. Outputs are depths below the datum, negative down.

The soundings file must hold depths below the datum. If the tide is added both to the file and through `TIDE`, it is counted twice; this happened in the Gulf composite runs and was fixed on 20 October 2026.

# 2 Lyzenga's regression

### 2.1 The model and its fit

With $X_m = \ln v_m$, the model (Lyzenga 1978, 1985; Lyzenga et al. 2006) is

$$\hat{Z} = a_0 + \sum_{m=1}^{n} a_m X_m = \mathbf{a}^{\mathsf{T}}\mathbf{x}, \qquad \mathbf{x} = (1, X_1, \ldots, X_n)^{\mathsf{T}}. \tag{2.1}$$

Setting `LSM 0` gives Lyzenga et al.'s $\ln L$ form. Fitted to every scene's bands at once, equation (2.1) is the *joint* model. It includes every weighted sum of per-scene fits as a special case. The scenes' tides are constants per scene, so they go into the coefficients, and the target is depth below the datum (`TIDE 0`).

The rows are the $N$ soundings whose cells see the bottom in every band. The coefficients minimise

$$\sum_{i=1}^{N} w_i \big(Z_i - \mathbf{a}^{\mathsf{T}}\mathbf{x}_i\big)^2, \tag{2.2}$$

solved by Householder QR of $W^{1/2}A$, which gives the triangular factor $R$ with $R^{\mathsf{T}}R = A^{\mathsf{T}}WA$. The weights are $w_i = w_i^0\,\psi_i$:

- $w_i^0$ is 1, or 1/(count in the sounding's depth bin) with `BALANCE`.
- $\psi_i$ are Huber weights, from iteratively reweighted least squares: $\psi_i = \min\big(1,\ k_H s/|e_i|\big)$, where $e_i$ is the residual, $s = 1.4826\,\mathrm{median}|e_i|$, and $k_H = 1.345$ (`HUBER`). The iterations start from the least-squares solution.

### 2.2 Rescaling against regression dilution (`POWER`)

Noise in the predictors shrinks the fitted depths towards the mean, so deep water reads too shallow. `POWER` uses classical calibration to correct this:

1. At the soundings, the fit's depths are modelled as $\hat{Z} = c\,Z^d$, fitted robustly in logs: $\ln\hat{Z} = \ln c + d\ln Z$, with the same Huber IRLS.
2. The law is inverted: $Z' = (\hat{Z}/c)^{1/d}$.

By default, the inverse is applied only beyond the crossover $\hat{Z}_\times = c^{1/(1-d)}$. That is where the fit is compressed into too little depth, so shallow depths keep the fit's smaller scatter. `POWER ALL` applies the inverse at every depth. At the Gulf, the joint fit gives $\hat{Z} = 1.081\,Z^{0.959}$, with the crossover at 6.9 m.

### 2.3 The depth error estimate (`ERROR`)

The `ERROR` grid is the $1\sigma$ prediction error of a new depth at cell $\mathbf{x}$. Write $W$ for the final weights, rescaled to average 1 ($\tilde{w}_i = N w_i / \sum_j w_j$). The residual variance of a sounding of average weight is

$$s^2 = \frac{1}{N-p}\sum_{i=1}^{N} \tilde{w}_i\, e_i^2, \qquad p = n+1. \tag{2.3}$$

The coefficients' covariance follows from the QR factor:

$$C = s^2\big(A^{\mathsf{T}}\tilde{W}A\big)^{-1} = s^2\,\frac{\sum_j w_j}{N}\,(R^{\mathsf{T}}R)^{-1}. \tag{2.4}$$

The prediction variance at a cell with predictors $\mathbf{x}$ is then

$$\sigma^2_{\hat{Z}}(\mathbf{x}) = s^2 + \mathbf{x}^{\mathsf{T}} C\, \mathbf{x}. \tag{2.5}$$

It has two parts: the scatter of one sounding about the model, and the uncertainty of the coefficients, which grows as $\mathbf{x}$ moves away from the soundings' centre.

Through `POWER`, a small error in $\hat{Z}$ scales by $\mathrm{d}Z'/\mathrm{d}\hat{Z} = Z'/(d\,\hat{Z})$. So where the rescaling applies,

$$\sigma_{Z'} = \sigma_{\hat{Z}}\, \frac{Z'}{d\,\hat{Z}}. \tag{2.6}$$

**What it represents.** Because Huber weighting is used, $s^2$ in (2.3) is a robust variance. Outlying soundings are down-weighted, so $s$ (0.70 m at the Gulf) describes the core of the residuals, not their tails (in-sample rms 1.86 m). The estimate assumes

- residuals independent between soundings, with one variance at every depth and place;
- the model's form is right.

It therefore cannot see:

- saturation where the bands stop seeing the bottom;
- spatially correlated error, for example from a change of bottom type;
- errors in the soundings themselves.

**Calibration at the Gulf.** This is the joint model, four bands per scene, with `POWER`, scored against ICESat-2 cells that the fit did not use (Table 1).

- **The core is conservative:** 80% of the errors lie within $\pm 1\sigma$, against 68% for a Gaussian.
- **The tails are heavy:** 93.6% lie within $\pm 2\sigma$, against 95%. The rms of $e/\sigma$ is 1.6.
- $\sigma$ **hardly varies:** it is 0.70–0.76 m at every depth, because $\mathbf{x}^{\mathsf{T}} C \mathbf{x}$ is small next to $s^2$ for 6,017 soundings.
- **It misses the saturation:** below about 16 m, the 20–30 m cells have no errors within $\pm 1\sigma$.

So it is a fair guide to the typical error in the depths the model reaches. It is not a bound, and it says nothing about saturation.

## 3 The attenuation behind the regression

Lyzenga's model is the linearisation of a two-flow model of the water-leaving signal over an optically shallow bottom:

$$L_m - L_{\infty,m} \;\approx\; A(\mathbf{x})\, r_m\, \exp\!\big(-g_m(\mathbf{x})\, Z_s\big), \qquad g_m = K_m\, M_s, \tag{3.1}$$

where

- $A(\mathbf{x})$ is the bottom's brightness, taken as spectrally flat in logs (Lyzenga's assumption);
- $r_m$ is the band's bottom colour, together with the band's deep-water and surface constants;

Table 1: Calibration of Lyzenga's `ERROR` on the Gulf's test half (joint model, four bands, `POWER`; 5,543 cells with a depth). Overall, 93.6% of the errors lie within $\pm 2\sigma$.

| Depth (m) | Cells | Within $\pm 1\sigma$ | Median $\sigma$ (m) | RMSE (m) |
|---|---|---|---|---|
| 0–2 | 161 | 83% | 0.70 | 2.38 |
| 2–5 | 1,352 | 87% | 0.70 | 1.32 |
| 5–10 | 2,027 | 79% | 0.74 | 0.78 |
| 10–20 | 1,980 | 77% | 0.75 | 0.97 |
| 20–30 | 23 | 0% | 0.76 | 8.77 |
| All | 5,543 | 80% | 0.74 | 1.21 |

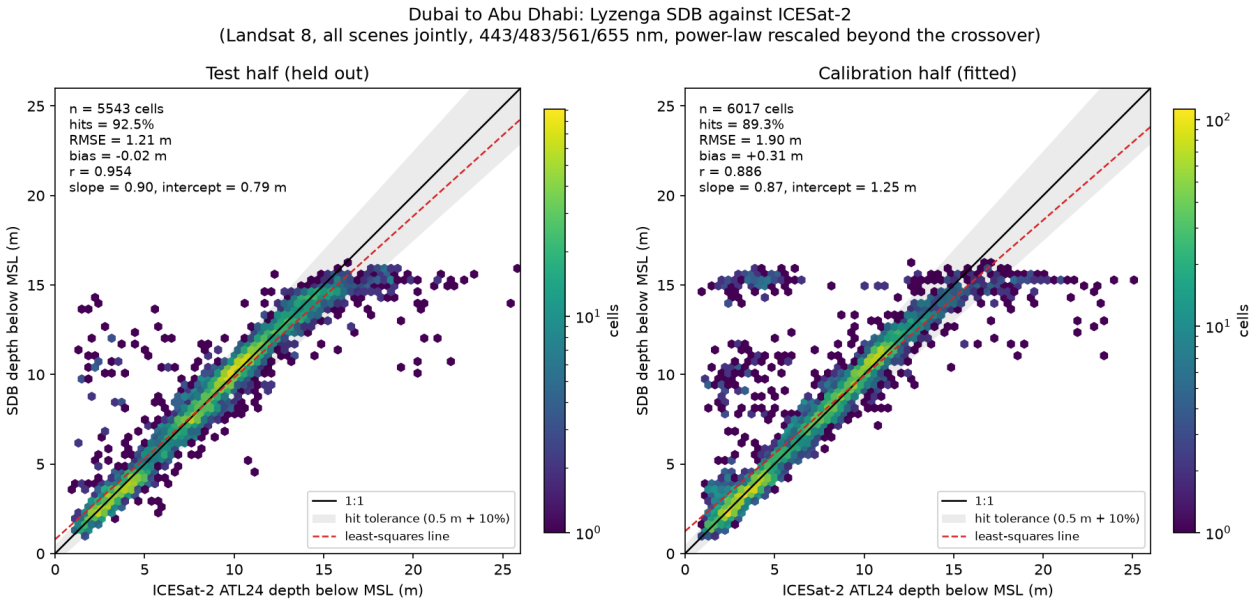



Figure 1: The joint Lyzenga model against ICESat-2 at the Gulf. The test half is held out by pass; the calibration half was fitted. Hits here count cells with a depth. Counting every reference cell, the test-half score is 90.0%.

- $K_m$ is the effective diffuse attenuation of the band;
- $M_s$ is the path length in water per metre of depth, down and back up.

$M_s$ comes from the sun and view zeniths in air (`SUNZENITH`, `VIEWZENITH`), refracted into water with $n_w = 1.34$:

$$M_s = \sec\theta_{w,\mathrm{sun}} + \sec\theta_{w,\mathrm{view}}, \qquad \sin\theta_w = \sin\theta / n_w. \tag{3.2}$$

With both zeniths at 0, $M_s = 2$. For the Gulf scenes (sun zenith 35–51°, near-nadir view), $M_s$ is about 2.1–2.25.

If the down- and up-welling attenuations differ, $g_m = K_d \sec\theta_{w,\mathrm{sun}} + K_u \sec\theta_{w,\mathrm{view}}$. The reported $K_m = g_m / M_s$ is then their path-weighted mean.

Taking logs of (3.1) gives, for grid $m$ of scene $s$,

$$X_m = \ln v_m = B(\mathbf{x}) + c_m - g_m(\mathbf{x})\, Z_s, \qquad B = \ln A,\ c_m = \ln r_m. \tag{3.3}$$

With $g_m$ and $c_m$ constant and $B$ constant, Lyzenga's regression (2.1) is this equation solved for $Z$. In practice $g_m$ varies over a scene, and between scenes, with the water. `ATTENUATION` estimates it explicitly, in two stages.

In stage 1, $L_{\infty,m}$ is the `LSM` value where that is positive, and otherwise the percentile. So `LSM 0` (Lyzenga et al.'s $\ln L$) still subtracts a deep-water signal for the attenuation.

# 4 Stage 1: the attenuation coefficients at the soundings

## 4.1 A smooth field for each grid

At the soundings the depth is known, so for each grid $m$ separately, equation (3.3) is fitted with $B$ left in the residuals:

$$X_{m,i} = c_m - g_m(\mathbf{x}_i)\, Z_{s,i} + \varepsilon_{m,i}. \tag{4.1}$$

$g_m(\mathbf{x})$ is a bilinear interpolant of values $\theta_{m,q}$ on a regular grid of nodes, set `NODES` degrees apart (default 0.05°; $12 \times 9$ nodes over the Gulf box):

$$g_m(\mathbf{x}) = \sum_{q=1}^{4} f_q(\mathbf{x})\, \theta_{m,q(\mathbf{x})}, \tag{4.2}$$

where $q(\mathbf{x})$ are the four nodes around the cell and $f_q$ their bilinear weights. The unknowns, $c_m$ and the $Q$ node values $\theta_{m,\cdot}$, minimise

$$\sum_i w_{m,i}\Big(X_{m,i} - c_m + Z_{s,i}\sum_q f_q(\mathbf{x}_i)\theta_{m,q}\Big)^2 \;+\; \lambda \sum_{(q,q')} \big(\theta_{m,q} - \theta_{m,q'}\big)^2. \tag{4.3}$$

The second sum runs over neighbouring pairs of nodes, east–west and north–south. It is a first-difference smoothing penalty, which makes the field smooth and fills nodes without soundings. Its weight is scaled to an average node's data:

$$\lambda = w_s\, \frac{\sum_i w^0_{m,i} Z^2_{s,i}}{Q}, \tag{4.4}$$

where $w_s$ is `SMOOTHING` (default 1; 0 requires soundings near every node). The normal equations, of size $Q+1$, are symmetric positive definite and are solved by Cholesky factorisation. The residuals are robustified by the same Huber IRLS as in Section 2, with scale $1.4826\,\mathrm{median}|\varepsilon|$.

## 4.2 Weights from the noise

The noise $\sigma_m$ on $v_m$ gives $\ln v_m$ a variance of about $\sigma_m^2/v_m^2$ (delta method). This grows without limit as the signal sinks towards the deep-water level. A floor $\epsilon = 0.05$ stands for the model's own error in $\ln v$. The weights are

$$w^0_{m,i} = \Big(\frac{\sigma_m^2}{v_{m,i}^2} + \epsilon^2\Big)^{-1}, \tag{4.5}$$

so a band counts fully while it sees the bottom clearly, and fades out as it loses it. The Huber factors multiply $w^0$.

### 4.3 Selecting soundings by the fitted signal

A sounding enters grid $m$'s fit only while its signal is above the noise, $v \geq k\sigma_m$. Selecting, and weighting, by the *observed* signal truncates the data on the dependent variable. At depth, only the cells that happen to be bright pass the test, which flattens the slope and makes $K$ too small. The effect is largest in the red, which loses the bottom within a few metres.

So after the first pass, each candidate sounding is reselected and reweighted by the *fitted* signal instead:

$$\hat{v}_{m,i} = \exp\big(c_m - g_m(\mathbf{x}_i)\, Z_{s,i}\big), \qquad \text{active if } \hat{v}_{m,i} \geq k\sigma_m, \tag{4.6}$$

with $\hat{v}$ in place of $v$ in (4.5). The fit is repeated until the selection stops changing, for at most 10 passes. This raised $K(655)$ at the Gulf by 20–40%. A grid needs at least 10 active soundings.

### 4.4 Outputs

- $K_m(\mathbf{x}) = g_m(\mathbf{x})/M_{s(m)}$ at every valid cell, as grid `/prefix/_/grid/`.
- `attenuation.csv`: $K$ of every grid at every node, with the node's support (the sum of the bilinear weights of the soundings near it).
- The printout gives, for each grid: $c_m$, the median $K$ over nodes with support of at least 1, its range, the robust scatter of (4.1), and the number of passes.

**Caveat.** $B$ is left in the residuals. $K$ is therefore unbiased only where the bottom's brightness does not change systematically with depth across the soundings. If seagrass is deeper than sand, the slope also contains the brightness change.

## 5 Stage 2: a depth from the attenuation

### 5.1 A straight line at every cell

With $c_m$ and $g_m(\mathbf{x})$ fixed from stage 1, equation (3.3) at a cell, over all its grids, is a straight line in $g_m$:

$$y_m \equiv X_m - c_m + g_m(\mathbf{x})\, h_{s(m)} = B(\mathbf{x}) - g_m(\mathbf{x})\, H(\mathbf{x}), \tag{5.1}$$

where $H$ is the depth below the datum. The intercept $B$ is the bottom's brightness and the slope is $-H$. The depth comes from how $g$ differs between bands and scenes, which is what makes it independent of $B$.

The weighted least-squares line through the grids that see the bottom gives

$$\hat{H} = -\frac{S_{gy}}{S_{gg}}, \qquad \hat{B} = \bar{y} + \hat{H}\bar{g}, \tag{5.2}$$

with

$$\bar{g} = \frac{\sum w_m g_m}{\sum w_m}, \quad \bar{y} = \frac{\sum w_m y_m}{\sum w_m}, \quad S_{gg} = \sum_m w_m (g_m - \bar{g})^2, \quad S_{gy} = \sum_m w_m (g_m - \bar{g})(y_m - \bar{y}). \tag{5.3}$$

The weights are (4.5) with the cell's signal. As in stage 1, the first pass selects and weights by the observed signal; later passes use the fitted signal,

$$\hat{v}_m = \exp\big(\hat{B} + c_m - g_m(\hat{H} + h_{s(m)})\big) = v_m \exp\big(\hat{B} - g_m\hat{H} - y_m\big), \tag{5.4}$$

until the selection settles (at most 10 passes). A cell needs at least 3 grids above $k\sigma_m$.

The outputs are three grids:

- `/prefix/_depth`, the depth $-\hat{H}$ (negative down);
- `/prefix/_depth_sigma`, its error $\sigma_H$;
- `/prefix/_bottom`, the bottom brightness $\hat{B}$.

### 5.2 The depth error estimate

$\sigma_H$ is the classical standard error of the slope of a weighted straight line, with the residual scale estimated from the cell's own scatter about the line:

$$\sigma_H = \sqrt{\frac{\mathrm{RSS}}{(n_u - 2)\, S_{gg}}}, \qquad \mathrm{RSS} = \sum_m w_m \big(y_m - \hat{B} + g_m \hat{H}\big)^2, \tag{5.5}$$

where $n_u$ is the number of grids used.

The leverage is $S_{gg}$, the weighted spread of the two-way attenuation among the cell's grids. It is small. Between 443 and 561 nm, $K$ differs by only 0.01–0.03 $\mathrm{m}^{-1}$, so $g$ spans about 0.02–0.07 $\mathrm{m}^{-1}$, and the scenes' different sun angles add little. A bottom colour that departs from spectral flatness by $\delta$ in ln therefore moves the depth by about $\delta/\Delta g$. For $\delta = 0.1$ that is 1.5–5 m, comparable to the depths being estimated.

Equation (5.5) treats the $n_u$ residuals as independent, with a common scale. It omits three things.

1. **Stage 1's uncertainty.** $c_m$ and $g_m(\mathbf{x})$ are taken as exact. Their errors are shared by every cell near the same nodes, so they are systematic, not random, and they do not average down.
2. **The bottom's colour.** A non-flat bottom spectrum gives each band its own offset in $y_m$. The same offset recurs in every scene, so the residuals are correlated across scenes, and with only a few distinct bands it can be fitted into the slope rather than showing up as scatter.
3. **Few effective degrees of freedom.** The 18–24 grids are only 3–4 distinct bands seen 6 times under similar water. $n_u - 2$ overstates the information.

All three make $\sigma_H$ too small. Out of fold at the Gulf, only 25% (four bands) and 37% (443, 483 and 561 nm) of the errors lie within $\pm\sigma_H$, against 68% for a calibrated estimate.

Two corrections are possible:

- propagate stage 1's covariance (the Cholesky factor is at hand) into (5.5);
- inflate $\sigma_H$ by a factor calibrated out of fold.

Neither is built.

## 6 Results at the Gulf

### 6.1 The attenuation coefficients

Over the box, $K$ is 0.05–0.09 $\mathrm{m}^{-1}$ at 443 nm, 0.07–0.10 at 483 and 0.065–0.105 at 561 (medians over the scenes), and it varies 1.5–2× across the box (Figure 3). The three bands are fitted independently, yet they map the same pattern: low in the south-west, rising towards Jebel Ali and Dubai. The scenes differ: 2019-03-13 is the clearest, 2018-10-04 the least clear. One set of Lyzenga coefficients cannot represent this variation. These maps are the per-site attenuation targets that the learned model's physics decoder needs (`validation/FUTURE_EXPERIMENTS.md`, H4(c)).

Pure water's absorption $a_w$ (Pope and Fry 1997) is a lower bound on $K$ (Figure 2).

- **443 and 483 nm:** $K$ is 4–15 times $a_w$, as expected where dissolved and particulate matter dominate.
- **561 nm:** $K$ is barely above $a_w$ (0.065–0.105 against 0.062 $\mathrm{m}^{-1}$). 6% of the node values fall below it, most of them in the clearest scene.
- **655 nm:** every value is below $a_w$ (0.08–0.19 against about 0.37 $\mathrm{m}^{-1}$), so none is physical.

The red loses the bottom within a few metres. Its deep-water signal varies (turbidity, adjacency, the atmospheric residual) by far more than the darkest cells' spread $\sigma_m$, which sets the threshold. So nearly every sounding stays active, and the slope is flattened. The flattening already reaches the green on clear scenes.

Two remedies are possible, and neither is built:

- fit the deep-water level per grid, or as a field;
- bound $K$ below by $a_w$.

A pure-water bound would bind at 561 nm as well as at 655.

### 6.2 The depth from the attenuation

Table 2 scores the depth from the attenuation against Lyzenga's depth, out of fold. The folds are the four ICESat-2 pass groups that chose Lyzenga's settings, within the calibration half only. The harness reproduces Lyzenga's cross-validation score (83.9%).

Table 2: Out of fold on the Gulf's calibration half (5,833 cells). A hit is an error within 0.5 m + 10% of the depth.

| | Hits | RMSE (m) | Bias (m) | 0–2 | 2–5 | 5–10 | 10–20 |
|---|---|---|---|---|---|---|---|
| Lyzenga, joint, four bands | 83.9% | 2.09 | +0.26 | 58% | 79% | 90% | 90% |
| Attenuation depth, four bands | 57.3% | 3.14 | −0.58 | 5% | 56% | 79% | 47% |
| Attenuation depth, 443/483/561 | 61.4% | 2.84 | +0.12 | 34% | 54% | 69% | 69% |

The depth from the attenuation is not competitive with Lyzenga's. With so little spread in $g$, a change of bottom type (seagrass against sand) shifts the band ratios enough to move the depth by metres. Lyzenga's regression absorbs the site's bottom–depth relationships into its coefficients; equation (5.1) cannot. Dropping the red removes the unphysical $K$ and helps the shallowest water most (0–2 m: 5 → 34%). Its error estimate is too small (Section 5.2).

So `ATTENUATION` is kept, off unless asked, with no effect on Lyzenga's fit. $K$ at 443–561 nm is the usable product. The attenuation depth is a diagnostic.

## 7 Usage

```
MODEL lyzenga_log_linear GRIDS g1 ... gn BATHYMETRY H [options, any order]
  LSM l1 ... ln | PERCENTILE p    DEEP k    RADIUS r    DEPTHS zmin zmax
  TIDE h    HUBER k    BALANCE bin    POWER [ALL]    ERROR H_error
  ATTENUATION K SCENES S [TIDES h1 ... hS] [SUNZENITH t1 ... tS]
                         [VIEWZENITH t1 ... tS] [NODES deg] [SMOOTHING w]
```

`SCENES` must come before the per-scene lists, and the grids are scene-major. The outputs are:

- `H` and `H_error`;
- with `ATTENUATION`, `K_/grid/` for every grid, and `K_depth`, `K_depth_sigma` and `K_bottom`;
- the files `model.bam` (the fitted model as applied), `output.csv` (every sounding, its model depth and whether it was used) and `attenuation.csv`.

At the Gulf, `validation/gulf_lyzenga.py` runs it. That script prepares the inputs, chooses the settings by cross-validation, scores the results and draws the figures; the modelling is all in *photic*. `make test` checks the fit and the attenuation on synthetic data that follow the models exactly:

- `tests/lyz_*`: the coefficients and every sounding's depth;
- `tests/att_*`: $K$ at every node, to $3 \times 10^{-6}$, and the depth at every cell, within 5 mm (the float precision of the band grids).

## References


- Lyzenga, D. R. (1978). Passive remote sensing techniques for mapping water depth and bottom features. *Applied Optics* 17(3), 379–383.
- Lyzenga, D. R. (1985). Shallow-water bathymetry using combined lidar and passive multispectral scanner data. *International Journal of Remote Sensing* 6(1), 115–125.
- Lyzenga, D. R., Malinas, N. P. and Tanis, F. J. (2006). Multispectral bathymetry using a simple physically based algorithm. *IEEE Transactions on Geoscience and Remote Sensing* 44(8), 2251–2259.
- Pope, R. M. and Fry, E. S. (1997). Absorption spectrum (380–700 nm) of pure water. II. Integrating cavity measurements. *Applied Optics* 36(33), 8710–8723.

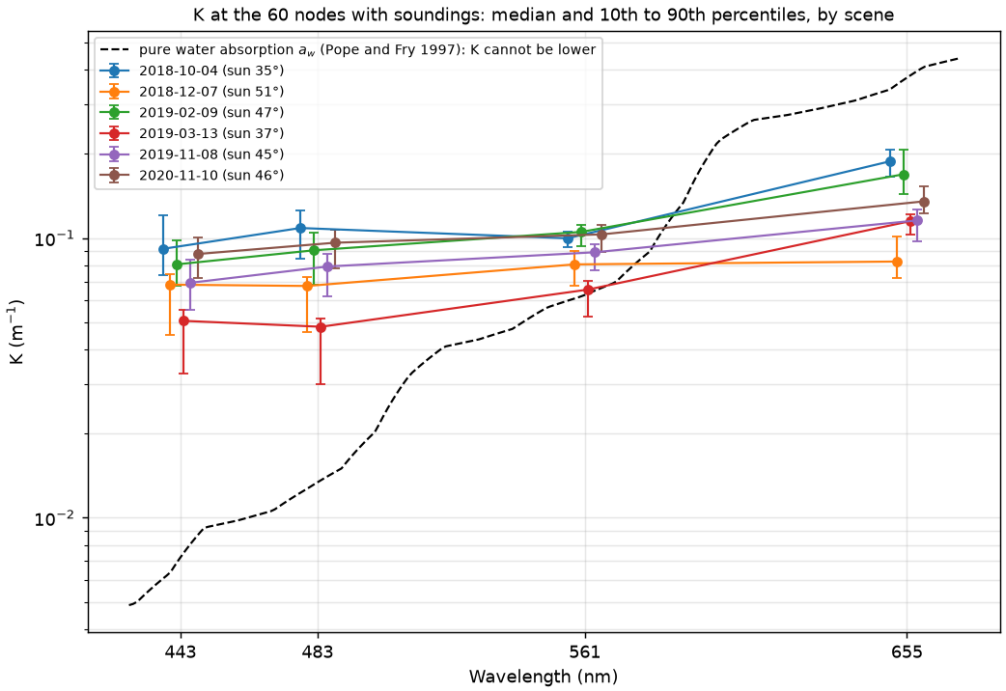



Figure 2: $K$ of each scene against wavelength at the 60 nodes with soundings (median, with the 10th to 90th percentiles), and pure water's absorption $a_w$. $K$ cannot physically fall below $a_w$.

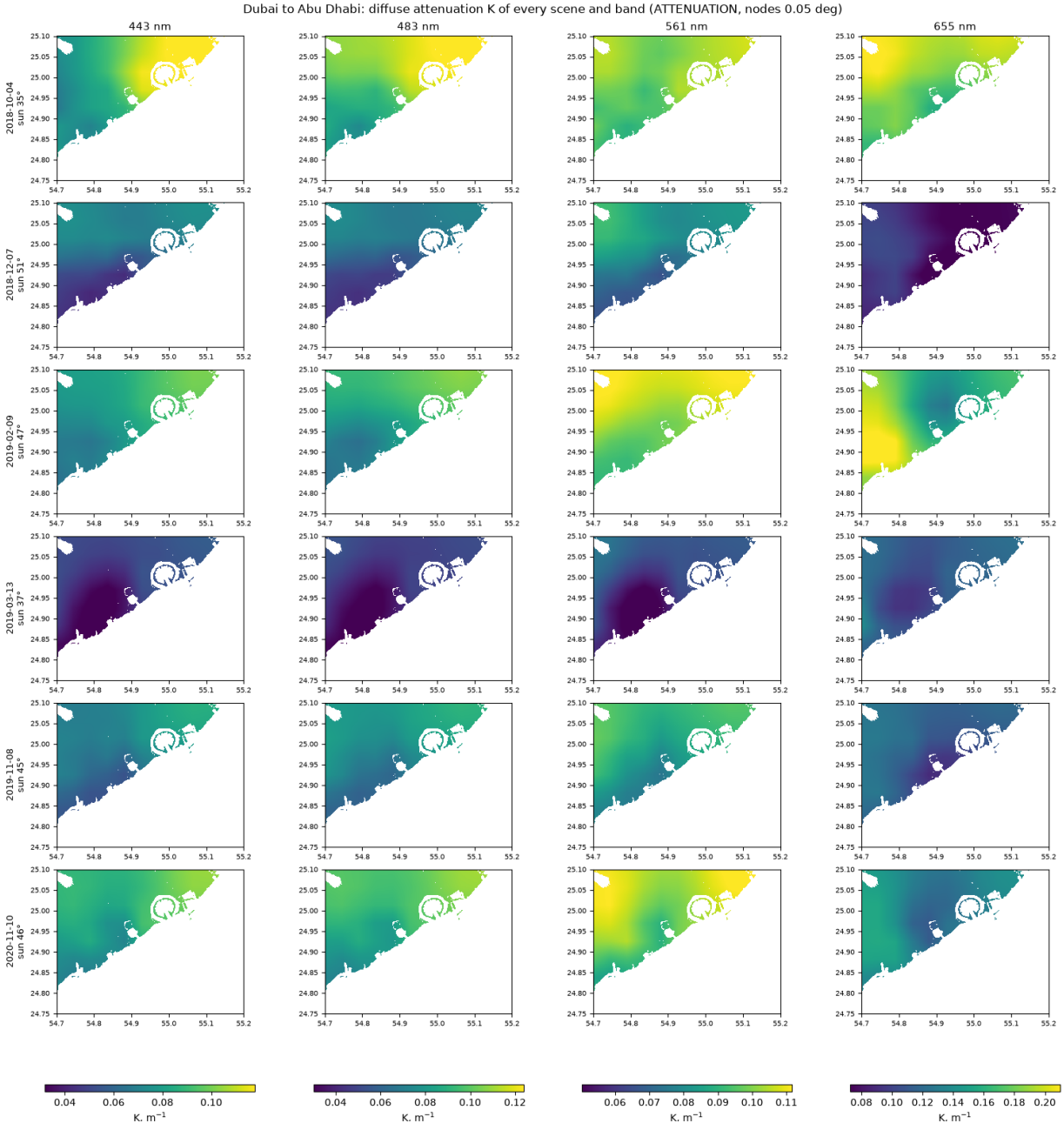



Figure 3: $K$ of every scene (rows, with the date and sun zenith) and band (columns), on one colour scale per band. Nodes are 0.05° apart.